26900241 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFTQ300 - FOOD ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Mention the main steps involved in the procedure of microwave ashing. [U] / [CO1]
2. Write the disadvantages of Phenol-Sulfuric Acid method ? [A] / [CO2]
3. Define oxidation and reduction. [K] / [CO3]
4. Mention any two limitations or interfering factors in protein estimation by UV absorption at 280 nm. [U] / [CO2]
5. What is the purpose of adding oxidizing agents during wet ashing? [K] / [CO1]
6. Name any two types of equipment used in the dry ashing method. [K] / [CO1]
7. Write down the method stated in Englyst-Cummings? [A] / [CO2]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Discuss the advantages, limitations, and precautions of the Gerber method for milk fat determination. [U] / [CO3]

9. Describe the color development reaction in the Somogyi-Nelson method. [U] / [CO2]

10. Explain the principle of Vitamin C estimation by the DCPIP method. [U] / [CO3]

11. Describe the construction, working principle, and applications of a hydrometer in food analysis. [U] / [CO1]

12. Describe the different steps involved in the Kjeldahl method: digestion, distillation, and titration. [U] / [CO2]

13. Differentiate between Mohr's method and Volhard's method. [U] / [CO3]

14. Explain the principle behind Karl-Fischer titration and add a note on KFReq [An] / [CO1]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Differentiate between crude fiber and dietary fiber. Explain their importance in food analysis. [An] / [CO2]

16. Compare simple distillation and reflux distillation with an immiscible solvent. Discuss their principles, procedures, advantages, limitations, and applications in food analysis, with suitable examples [U] / [CO1]

17. Evaluate the importance of determining soluble, insoluble, and total dietary fiber in food and nutritional analysis. [E] / [CO2]

18. Write a detailed essay on the Modified Mojonnier method as a discontinuous solvent extraction technique for milk fat estimation. Include its principle, characteristics, step-by-step procedure, precautions, advantages, limitations, and applications in dairy and food analysis. [A] / [CO3]

[10x2 = 20]